Federated Learning for Pneumonia Classification using GAN-Augmented Chest X-Rays

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***Abstract*—AI-driven solutions face several challenges when applied to the healthcare field. Strict data privacy regulations, the lack of a unified data source, and the unbalanced nature of medical datasets (which underrepresent some diagnostic classes) are primary barriers. This report presents a framework that tackles these issues by applying GAN-based dataset augmentation, combined with a Federated Learning (FL) pipeline, to train a CNN to classify pneumonia in Chest X-Rays. This specific use case was selected for its simplicity, as the main goal of the project is to illustrate the viability of the end-to-end distributed framework. Results indicate the framework successfully equilibrates the dataset and maintains robust diagnostic performance, achieving a Federated Test AUC of 90% without centralizing sensitive raw data.**

I. Introduction

The use of Artificial Intelligence (AI) in healthcare environment presents exceptional opportunities to optimize processes, improve diagnosis precision and relieve healthcare staff from executing repetitive tasks to provide a better treatment to patients. Unlike other machine learning applications, medical data cannot be easily centralized for training models, as strict privacy policies are imposed to protect the patient. Furthermore, clinical information is inherently distributed across multiple distinct hospitals and organizations, and the datasets themselves are frequently unbalanced due to the scarcity of specific diagnostic classes or rare diseases.

This mini-project illustrates these core challenges and proposes a modular, end-to-end AI pipeline designed to overcome them. To validate the architecture, we selected the binary classification of Chest X-Rays (Pneumonia vs. Normal). While modern Convolutional Neural Networks (CNNs) can easily achieve high accuracy on this specific task, the dataset presents a significant class imbalance due to a scarcity of the minority 'NORMAL' class, making it an ideal candidate for our project.

The true goal of this report is to demonstrate a framework that navigates privacy, distribution, and data scarcity. To achieve this, the proposed system architecture leverages three distinct techniques:

- **Generative Adversarial Networks (GAN):** A Deep Convolutional GAN (DCGAN) is implemented to generate realistic synthetic images and augment the minority class ('NORMAL' class in our use case).
- **Convolutional Neural Networks (CNN):** Performs the binary diagnostic classification (Pneumonia vs. Normal).
- **Federated Learning (FL)**: Enables model training across simulated hospital clients without pooling data. Only model weights are shared centrally via `FedAvg`, keeping raw data local.

The report is organized as follows: Section II details the system architecture, including the GAN, CNN, and Federated Learning configurations. Section III presents the comparative results between centralized and federated training. Finally, Section IV outlines the main conclusions.

II. Methodology and System Architecture

A. DCGAN Synthetic Dataset Generation

A standard DCGAN architecture [1], based on the official TensorFlow implementation[1], was used to equilibrate the dataset, was used to equilibrate the dataset, augmenting the samples of the `NORMAL` minority class.

The DCGAN architecture is composed of two adversarial neural networks:

- **Generator:** The network samples a 100-dimensional latent noise vector (100×1) and processes it through a Dense layer and a Reshape operation to form an initial $8 \times 8 \times 256$ feature map. This is followed by a sequence of `Conv2DTranspose` layers with fractional strides to upsample the feature maps to $128 \times 128 \times 1$. Batch Normalization and `LeakyReLU` activations are applied after each transposed convolution, concluding with a `tanh` activation function for the final image generation.
- **Discriminator:** The discriminator evaluates the realism of the input images ($128 \times 128 \times 1$). It utilizes a series of standard `Conv2D` downsampling layers (filter depths of 64, 128, 256, and 512), coupled with `LeakyReLU` activations and Dropout regularization (rate of 0.3). The final layer is flattened and mapped to a single Dense unit that outputs a scalar real/fake score.

The model architecture is illustrated in the Figure 1. Both networks were compiled using Binary Crossentropy loss and trained using Adam optimizers with a learning rate of 10^{-4}. The training cycle was executed over 7000 epochs.

B. CNN Classifier Architecture

The CNN takes 128×128 grayscale chest X-rays and classifies them as pneumonia or normal. Its design is inspired by VGG-style [2] networks, using four convolutional blocks with 3×3 convolutions, batch normalization, LeakyReLU, and max pooling. The number of filters doubles after each block

[1] https://www.tensorflow.org/tutorials/generative/dcgan

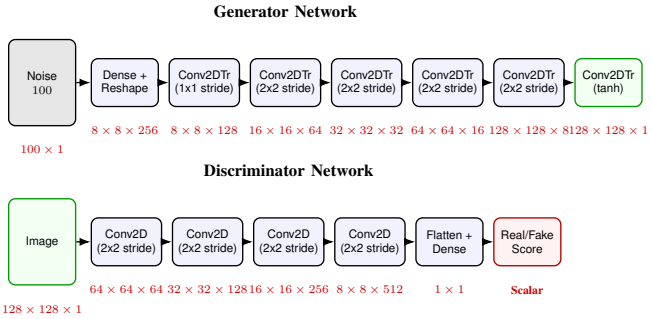


Fig. 1: DCGAN Architecture: The Generator maps latent space noise to 128×128 X-Ray scans, while the Discriminator evaluates realism via downsampling.

from 32 up to 256. This structure lets the model learn from simple to more complex image features.

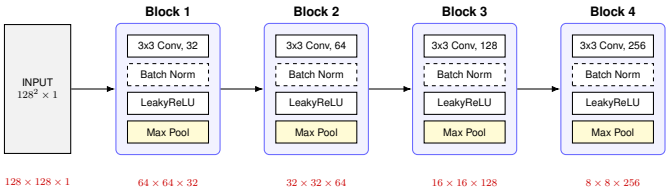


Fig. 2: Feature extraction backbone of the CNN Architecture.

After the convolutional blocks, the model uses global average pooling to turn the feature maps into a 256-element vector, as shown in Figure 3. This keeps the classification head smaller than a flat dense layer would. The vector goes through a dense layer with 128 units and ReLU, then dropout at 0.5. The final output is a single sigmoid neuron that gives the pneumonia probability.

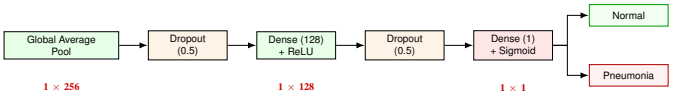


Fig. 3: Classification head of the CNN Architecture.

Model optimization is driven by the Adam optimizer initialized with a conservative learning rate of 3×10^{-5} to ensure steady convergence across the loss surface. Given the binary structure of the objective function, training leverages a Binary Cross-Entropy loss mechanism, complemented by a `ReduceLROnPlateau` scheduling policy designed to dynamically downgrade the learning rate whenever validation loss plateaus. The centralized baseline model was trained for a total of 50 epochs. Because the source medical imaging data exhibits an uneven class distribution, evaluation protocols look past baseline accuracy alone. Instead, model stability and generalization are contextualized comprehensively through Area Under the ROC Curve (AUC), Precision, Recall, and F1-Score metrics to guarantee performance validity under distinct class imbalances.

C. Federated Learning Pipeline

The Federated Learning pipeline enables collaborative model training across multiple nodes without sharing their raw data, ensuring data privacy and distributing the computational load. The pipeline was implemented using the *Flower* framework [3] and its *Virtual Client Engine (VCE)*, which enables experimentation with multiple federation setups. In this mini-project, the simulation setup was configured as follows:

- **Clients:** 4 independent nodes.
- **Data split:** Randomly distributed (Independent and Identically Distributed (IID), 25% of the dataset per client).
- **Training:**
  - **Local Execution:** 1 epoch per client per round.
  - **Global Execution:** 50 total communication rounds (equivalent to the centralized baseline epochs).
  - **Synchronization:** Synchronous updates (nodes wait for all others to finish before starting the next round).
- **Model Aggregation Strategy:** `FedAvg`

A single local execution epoch was configured to avoid local model overfitting, which commonly occurs when a high number of local epochs are executed within the federation. This ensures that one federation round is equivalent to one centralized epoch in terms of data passes through the model. The standard `FedAvg` aggregation algorithm was used to perform the global model weights update.

III. Results and Evaluation

A. GAN Evaluation

To quantitatively evaluate the quality and realism of the synthetic chest X-ray images generated by the DCGAN, the Fréchet Inception Distance (FID) [4] was used as the primary metric. The FID computes the distance between the feature distributions of real and synthetic images, where a lower score designates a higher similarity to the real dataset.

Upon completing the 7000 training epochs, the generative model achieved a final FID score of 25.37. This outcome indicates a successful generation of synthetic data that accurately reflects the structural properties of genuine patient scans. By demonstrating this level of visual fidelity, the augmented images were validated for use in equilibrating the dataset. This successfully augmented the `NORMAL` minority class, overcoming the initial data imbalance and providing a robust foundation for the downstream classification tasks. Furthermore, this data augmentation framework is highly extensible and can be adapted to broader medical domains requiring dataset equilibration.

B. Classification Performance

The CNN classifier was evaluated under both a centralized baseline and a federated training regime. In the centralized setting, the model reached a test accuracy of 83.7%, an AUC of 77%, and a recall of 93%. Under federated learning, the classifier achieved 81% accuracy, a higher AUC of 90%, and a recall of 92%.

These outcomes show that federated training can preserve strong clinical sensitivity while improving the model's ability to distinguish pneumonia from normal cases. The federated training performance is likely better due to the identical distribution of data (IID), which typically makes model training through federation simpler. More experiments would be needed using different data splits to conclude if FL can *always* be equivalent to centralized training in model convergence. The higher AUC for the federated model indicates better overall ranking performance, even when accuracy remains slightly lower than the centralized baseline.

High recall is particularly important for pneumonia screening, and both approaches maintain this property. The federated model therefore represents a viable alternative for distributed medical classification systems, balancing privacy-preserving training with competitive diagnostic performance.

IV. Final Metrics

Summary of final quantitative outcomes from the project. The report follows the DCGAN-based augmentation approach and evaluates image quality with Fréchet Inception Distance (FID) [1], [4].

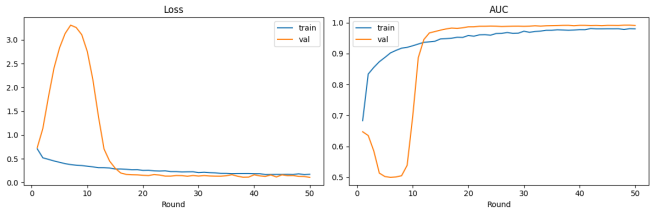


Fig. 4: Federated: train/val loss and AUC across rounds.

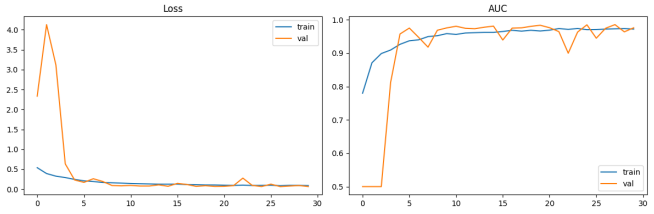


Fig. 5: Per-client training curves (loss and AUC).

Metric	Value
GAN FID	25.37
Eval AUC (start)	0.647
Eval AUC (end)	0.991
Best eval loss	0.111

TABLE I: Selected final metrics (rounded).

V. Conclusions

This project shows that a practical pipeline for pneumonia classification can be built from two complementary components. First a DCGAN [1] was trained to augment the minority `NORMAL` class. The generated set reduced class imbalance and achieved a measured FID around 25 [4] which indicates reasonable visual fidelity for downstream use.

Second a compact CNN classifier trained in a federated setup [5] reached high discrimination on the evaluation set. Over 50 communication rounds the evaluation AUC stabilized near 0.99 and the evaluation loss reached values near 0.11. These results demonstrate that federated training can match centralized performance while keeping data local to simulated clients in IID data split scenarios.

Taken together the GAN augmentation and the federated CNN form a practical approach for sensitive medical imaging tasks. The pipeline improves data availability and preserves privacy while delivering strong classification metrics. At the same time there remain clear directions for improvement. The FID can be lowered by longer GAN training and richer data augmentation. The classifier would benefit from additional cross site validation and adversarial robustness checks. Finally real world deployment requires formal privacy auditing and fairness analysis across demographic groups.

Overall the work validates the feasibility of combining synthetic data generation with federated learning for medical image classification. Figures 4 and 5 show the global and per client training behaviour reported in the experiments. The experiments provide a reproducible baseline and a set of artifacts to continue development. Numeric logs are available in ‘results/federated_metrics.csv‘ and ‘fid_score_results.txt‘ in this repository. The full code and generated artifacts are published on the project GitHub repository https://github.com/cyberpabs/FedXR-GAN.

References

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